

Chemistry Exam-02 — Attempt Overview

Chapters 3, 4, 5 · Atomic Structure, Periodic Table, Chemical Bonding · Seemab · 2026-04-17

59.5 / 65

91.5% · Grade A+

12 / 12

Section A — MCQs

100%

30.5 / 33

Section B — Short

92.4%

17 / 20

Section C — Long

85%

What Went Well

- **MCQs — perfect 12/12.** Every concept nailed: atomic number, neutrons in ^{37}Cl , electronegativity, periodic trends, coordinate bonds, hydrogen bonding. No random guessing.
- **Dalton's atomic theory (Q.2(i))** — all 4 postulates + importance, full marks.
- **Hydrogen isotopes (Q.2(iii))** — clean tabular answer with symbols and neutron counts.
- **Hydrogen bonding in water (Q.2(xi))** — went beyond the definition to link to water's unique properties (boiling point, surface tension, universal solvent).
- **Covalent bond formation (Q.3(iv))** — perfect 5/5. All three bond types (single H_2 , double O_2 , triple N_2) explained with clear electron-dot representation.
- **Tabular organization** throughout Section B and C — easy-to-read answers.

Where Marks Were Lost

- **Atomic size trend down a group (Q.3(iii), -1 mark)** — wrote "decreases" but correct answer is **increases**. The reasoning she gave ("new shells added") actually supports "increases" — a careless conclusion, not a conceptual error.
- **Ionization energy down a group (Q.2(vi), -1 mark)** — sub-question left blank. She wrote the heading but never completed the sentence.
- **Element labels mixed up (Q.2(iv), -1 mark)** — correct configurations for all three elements (Na, Al, Cl), but labels were swapped between Al and Cl.
- **Isotope "uses" missing (Q.3(ii), -1 mark)** — described the isotopes thoroughly but forgot to mention their uses (C-14 dating, deuterium heavy water, etc.)
- **Rutherford's 3rd defect missing (Q.3(i), -1 mark)** — gave 2 of 3 defects. Missing: couldn't explain atomic stability / multi-electron atoms.
- **$\text{Cl}=\text{Cl}$ instead of $\text{Cl}-\text{Cl}$ (Q.2(x), -0.5 mark)** — chlorine gas has a single bond. Small error; everything else in that question was correct.

Patterns to Watch

The errors on this exam share a common theme: **rushed endings**. MCQs (done first, carefully) were perfect. Short answers (middle of the paper) had a few slips. Long answers (end of paper, under time pressure) is where most marks leaked out — incomplete ionization-energy trend, missing Rutherford defect, missing isotope uses. **Practice ending each answer with a "have I covered all parts?" mental check.**

Specific Revision Targets (before next exam)

Topic	Source
Periodic trends — direction for <i>each</i> of size, IE, EA, EN across period and down group	Unit 4 (Periodic Table), explanations.md + flashcards
Rutherford's model — all 3 defects	Unit 3 (Atomic Structure), explanations.md
Isotope uses (C-14, deuterium, I-131, Co-60, etc.)	Unit 3, stories-and-examples.md
Bond types in diatomic halogens (Cl ₂ , Br ₂ , I ₂ all single)	Unit 5, notes.md

Position in Track Record

This is Seemab's **5th exam** so far. Prior scores:

Exam	Subject	Chapters	Score	%
Exam-01	Biology	Ch 1-6	52 / 65	80%
Exam-01	Physics	Ch 1-2	63.5 / 65	97.7%
Exam-02	Physics	Ch 3-4, 9	62 / 65	95.4%
Exam-01	Chemistry	Ch 1-2	64 / 65	98.5%
Exam-02	Chemistry	Ch 3-5	59.5 / 65	91.5%

Running average: 92.6% (up from 92.9% before — slight dip because this exam scored below the previous Chem exam-01's 98.5%).

Report date: 2026-04-17 · **Attempt source:** chemistry/exams/exam-02-ch3-5/attempts/2026-04-17/scans.pdf · **Detailed marking:** same dir/report.pdf